



 Alfaisal University

# LLM Augmented

IoT Traffic Management System

**PRESENTED BY**

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# The Problem

Traffic congestion at signalized intersections remains a **critical urban issue**.

- Adaptive controllers lack user-friendly interfaces.
- Technical operators are often unavailable in smaller installations.
- There's a significant gap between intelligent hardware and human usability.
- A safe, natural language control system is essential.



# System Architecture

## LAYER 1: ESP32

The first layer utilizes the ESP32 microcontroller equipped with ultrasonic sensors, providing real-time data on traffic conditions while ensuring minimal invasiveness to existing infrastructure.

## LAYER 3: FASTAPI BRIDGE

In the third layer, the FastAPI bridge acts as an intermediary, processing and managing data with a rolling state window for enhanced responsiveness and streamlined operations.

## LAYER 2: MQTT BROKER

The second layer incorporates the HiveMQ MQTT broker for efficient telemetry, facilitating reliable communication between devices and ensuring smooth data flow throughout the system architecture.

## LAYER 4: CLAUDE API

Layer four features the Claude API, which enables constrained tool use, ensuring safety and reliability while processing commands, minimizing the risk of errors and unauthorized operations.

# Advantages of Constrained Tool Use

## STRUCTURAL SAFETY

Constrained tool use guarantees **structural safety** by design, effectively minimizing the risks associated with unexpected behaviors or errors during operation, ensuring a reliable traffic management system.

## VALID PARAMETERS

By eliminating hallucinated parameters, the system prevents **malformed JSON output**, which can cause failures in communication and data processing within the IoT traffic management framework, enhancing overall reliability.

## GUARANTEED VALIDITY

Every tool call is guaranteed valid, which reinforces the **schema enforcement at the platform level**, maintaining operational integrity even in varying states of firmware or network conditions.

# Implementation Highlights of the System

## OVERLAY INTEGRATION

The system allows for **seamless overlay integration** above the existing firmware, ensuring the original operation remains unaffected while enhancing functionality with new capabilities.

## ADAPTIVE TIMING

Our firmware implements a **four-phase adaptive timing** algorithm, which supports dynamic adjustments based on real-time traffic conditions, optimizing flow through signalized intersections effectively.

## TOOL UTILIZATION

The integration features **single-turn tool use**, including two specialized tools: `traffic_override` and `emergency_priority`, enabling precise control while maintaining all parameters within a defined schema.

# Evaluation Methodology Overview

## DATASET COMPOSITION

The evaluation relies on a **labelled dataset** of 40 operator utterances categorized into four distinct groups, ensuring comprehensive analysis across various communication scenarios.

## CATEGORY BREAKDOWN

Each category consists of **10 utterances**, including Command, Ambiguous, Out of Scope, and Reporting, enabling a nuanced understanding of operator interaction and system response.

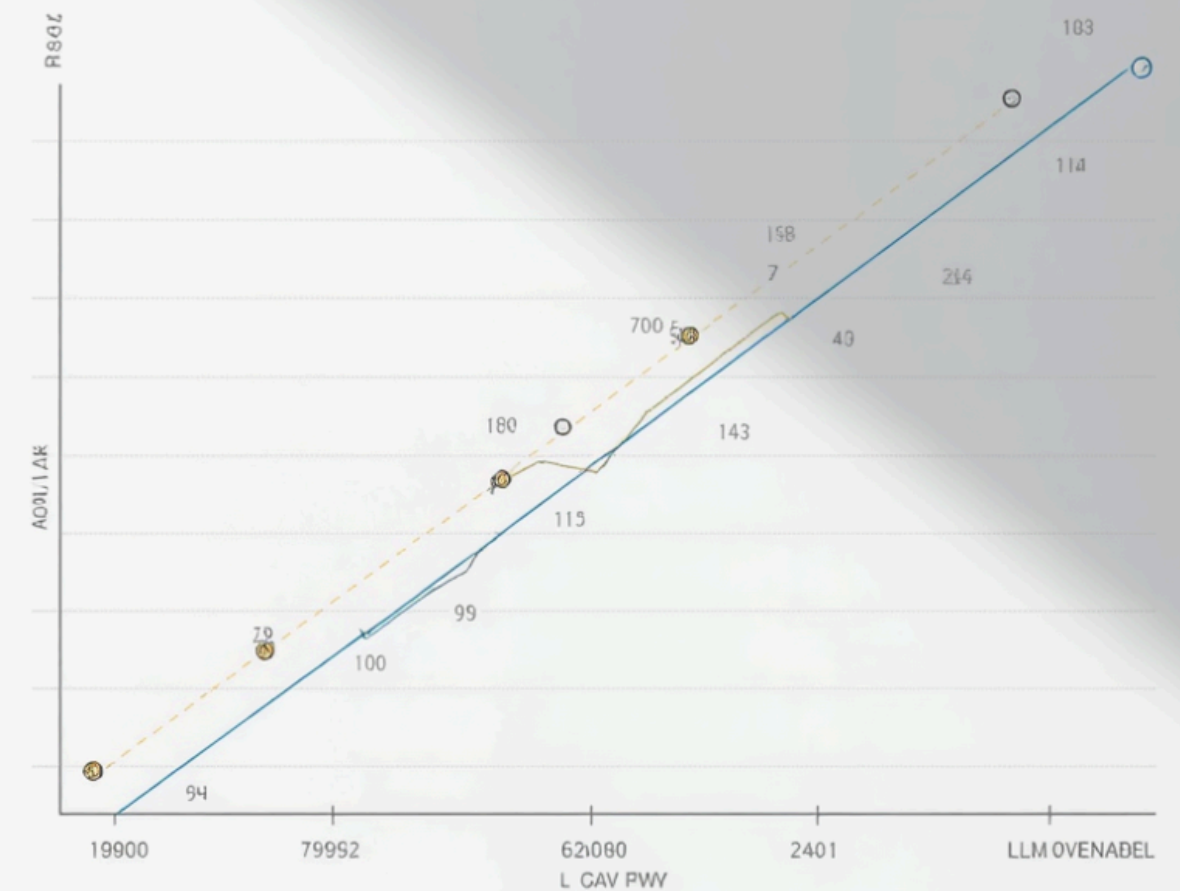
## BASELINE COMPARISON

A deterministic regex matcher serves as a **baseline**, achieving 75% accuracy, providing a crucial reference point for evaluating the performance of the developed LLM-augmented system.

# 97.5%

## LLM Overlay Accuracy Rate

The LLM overlay **outperformed** the baseline model, achieving an impressive **97.5% accuracy**, demonstrating significant improvements in tool detection and structural validity during traffic management operations.





# Contributions to Traffic Management Research

## NON-INVASIVE OVERLAY

The system employs a **non-invasive overlay** on existing firmware, providing a seamless integration that requires no modifications, thereby enhancing compatibility and reducing deployment complexity.

## UNIFIED ENDPOINT

A **single unified endpoint** is established for both command and reporting functionalities, streamlining interactions and enabling efficient communication between components without increasing system overhead.

## REPRODUCIBLE EVALUATION

The use of a **stratified dataset** for evaluation ensures reproducibility and transparency, allowing for consistent benchmarking and assessment of system performance across different scenarios.



# Future Directions

## DEPLOYABLE SYSTEM

The objective is to transition from the reference design to a fully deployable system that effectively manages urban traffic and adapts to real-time conditions.

## LARGER DATASET

Sourcing a larger evaluation dataset externally will improve the accuracy and reliability of the system, facilitating robust performance assessments across diverse traffic scenarios.

## LOCAL SLM

Implementing a local SLM will enhance system responsiveness, reducing latency while ensuring user privacy is maintained through localized data processing strategies.

## MULTI-INTERSECTION COORDINATION

Developing multi-intersection coordination scenarios will allow for optimized traffic flow management, enhancing overall urban mobility and reducing congestion levels in critical areas.